

# Vertex-Fault-Tolerant Hamiltonicity of Cube-Connected Cycles Graphs \*

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## Abstract

The graph  $CCC_n$  has  $n \times 2^n$  vertices, labeled  $(l, \mathbf{x})$ , where  $l$  is an integer between 0 and  $n - 1$  and  $\mathbf{x}$  is an  $n$ -bit binary string. Two vertices  $(l, \mathbf{x})$  and  $(l', \mathbf{y})$  are adjacent if and only if either  $\mathbf{x} = \mathbf{y}$  and  $|l - l'| = 1$ , or  $l = l'$  and  $\mathbf{y} = (\mathbf{x})^l$ . A graph  $G$  is called  $k$ -vertex-fault-tolerant hamiltonian if it remains hamiltonian after removing at most  $k$  vertices. In this paper, we will prove that  $CCC_n$  is 1-vertex-fault-tolerant hamiltonian if  $n$  is odd and  $n \geq 3$ .

**Keywords:** hamiltonian, cube-connected cycles network, vertex-fault-tolerant hamiltonian.

## 1 Introduction

For the graph definition and notation, we follow [7]. The cube-connected cycle  $CCC_n$  was introduced by Preparata and Vuillemin in 1981 [8] as a good alternate for the hypercube, having a fixed degree equal to 3 and yet a small diameter compared to its number of vertices. This graph was since then studied by a lot of people, who showed in particular, that it also has good properties as far as communications are concerned. It was proved that  $CCC_n$  contains a hamiltonian cycle, which is

an interesting property to realize distributed algorithms. However, its cycle structure is still not completely known, as to whether it contains cycles of any given length. Germa, Heydemann, and Sotteau [5] further study the existence of cycles of all lengths in the cube-connected cycles graph and obtain the following theorem.

**Theorem 1.1** [5] *The cube-connected cycles graph  $CCC_3$  contains cycles of length 3 and all lengths between 8 and 24.  $CCC_4$  contains cycles of length 4 and cycles of all even lengths between 8 and 64. For  $n \geq 5$ ,  $CCC_n$  contains cycles of length  $n$  and of every even length between 8 and  $n2^n$  except 10 and possibly  $n2^n - 2$ . Furthermore, for  $n$  odd,  $CCC_n$  contains cycles of every odd length between  $n + 6$  and  $n2^n - n - 2$  and also cycles of length  $n2^n - n + 2$ .*

In this paper, we only concentrate on the existence of cycle of length  $n2^n - 1$  in  $CCC_n$ . It is proved that  $CCC_n$  is a bipartite graph if and only if  $n$  is even. Obviously, there is no cycle of length  $n2^n - 1$  in  $CCC_n$  if  $n$  is even. We will prove that there is a cycle of length  $n2^n - 1$  in  $CCC_n$  if  $n$  is odd and  $n \geq 3$ .

More formally, we say a graph  $G$  is called  $k$ -vertex-fault-tolerant hamiltonian if it remains hamiltonian after removing at most  $k$  vertices. Obviously, a  $k$ -vertex-fault-tolerant hamiltonian graph  $G$  with  $k \leq \delta(G) - 2$ , where  $\delta(G) = \min\{\deg(v) \mid v \in V(G)\}$ . In this paper, we will

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prove that  $CCC_n$  is 1-vertex-fault-tolerant hamiltonian if  $n$  is odd and  $n \geq 3$ .

## 2 Notation and definition

Let  $\mathbf{u} = u_{n-1}u_{n-2}\dots u_1u_0$  be an  $n$ -bit binary string. We use  $(\mathbf{u})_j$  to denote the bit  $u_j$  and  $(\mathbf{u})^k$  to denote the vertex  $\mathbf{v} = v_{n-1}v_{n-2}\dots v_1v_0$  with  $(\mathbf{u})_i = (\mathbf{v})_i$  for  $0 \leq i \neq k \leq n-1$  and  $(\mathbf{v})_k = 1 - (\mathbf{u})_k$ . Let  $\mathbf{0}^n$  denote the zero  $n$ -bit binary string with  $(\mathbf{0})_i = 0$  for  $0 \leq i \leq n-1$ .

The graph  $CCC_n$  has  $n \times 2^n$  vertices, labeled  $(l, \mathbf{x})$ , where  $l$  is an integer between 0 and  $n-1$ , called the *level* of the vertex, and  $\mathbf{x}$  is an  $n$ -bit binary string, called the *row* of the vertex. Two vertices  $(l, \mathbf{x})$  and  $(l', \mathbf{y})$  are adjacent if and only if either  $\mathbf{x} = \mathbf{y}$  and  $|l - l'| = 1$ , or  $l = l'$  and  $\mathbf{y} = (\mathbf{x})^l$ . In this last case,  $\mathbf{x}$  and  $\mathbf{y}$  differ in exactly the bit in position  $l$ . The *level*  $l$  of  $CCC_n$ ,  $Level(l)$ , is the set of all vertices with level  $l$ . The edges joining two vertices of the same level are called hypercube-edges. The *row*  $\mathbf{x}$  of  $CCC_n$ ,  $Row(\mathbf{x})$ , is the set of all vertices with row  $\mathbf{x}$ . The edges that connect  $(l, \mathbf{x})$  to its neighbors  $(l+1, \mathbf{x})$  and  $(l-1, \mathbf{x})$  are called cycle-edges. Obviously, when  $n \geq 3$ ,  $Row(\mathbf{x})$  forms a cycle of length  $n$  called the *fundamental cycle* defined by  $\mathbf{x}$ . For  $n = 2$ ,  $CCC_n$  is simply the cycle of length 8. So in this article we will only consider the case  $n \geq 3$ .

Let  $a \in \{0, 1\}$ . We set  $CCC_n^a$  denote the subgraph of  $CCC_n$  induced by the vertex set  $\{(l, \mathbf{x}) \mid 0 \leq l < n \text{ and } (\mathbf{x})_{n-1} = a\}$ . Obviously,  $\deg_{CCC_n^a}((n-1, \mathbf{x})) = 2$  for any  $\mathbf{x}$  with  $(\mathbf{x})_{n-1} = a$ . We replace the path  $\langle (n-2, \mathbf{x}), (n-1, \mathbf{x}), (0, \mathbf{x}) \rangle$  with  $\langle (n-2, \mathbf{x}), (0, \mathbf{x}) \rangle$  for any  $\mathbf{x}$  with  $(\mathbf{x})_{n-1} = a$ . Let the resultant graph is  $(CCC_n^a)'$ . It is easy to check that  $(CCC_n^a)'$  is isomorphic to  $CCC_{n-1}$  for every  $a \in \{0, 1\}$ .

With the above observation, we recursively construct  $CCC_n$  from four copies of  $CCC_{n-2}$  through the following three steps.

Step 1: Let  $ab \in \{00, 01, 10, 11\}$ . We set  $(CCC_n^{ab})'$  be the copy of  $CCC_{n-2}$  such that the vertex  $(l, \mathbf{x})$  in  $(CCC_n^{ab})'$  correspond to the vertex  $(l, \mathbf{y})$  in  $CCC_{n-2}$  satisfying  $(\mathbf{x})_{n-1} = a$ ,  $(\mathbf{x})_{n-2} = b$ , and  $(\mathbf{x})_i = (\mathbf{y})_i$  for  $0 \leq i \leq n-3$ . In other words, adding two more bits  $ab$  to each vertex of  $CCC_{n-2}$ .

Step 2: Insert all the level  $n-2$  and level  $n-1$  vertices. In other words, we replace the path  $\langle (n-3, \mathbf{x}), (0, \mathbf{x}) \rangle$  with the path  $\langle (n-3, \mathbf{x}), (n-2, \mathbf{x}), (n-1, \mathbf{x}), (0, \mathbf{x}) \rangle$ . Obviously, the subgraph induced by those vertices with  $(\mathbf{x})_{n-1} = a$  and  $(\mathbf{x})_{n-2} = b$  is  $CCC_n^{ab}$ .

Step 3: Adding all level  $n-2$  and level  $n-1$  edges to connected  $CCC_n^{00}$ ,  $CCC_n^{01}$ ,  $CCC_n^{10}$ , and  $CCC_n^{11}$ . The resultant graph is  $CCC_n$ .

With this observation, we introduce the following notation. For each subgraph  $G$  of  $CCC_{n-2}$ , we use  $G^{ab}$  to denote the corresponding subgraph graph in  $CCC_n^{ab}$ . We use  $(G^{ab})'$  to denote the corresponding subgraph graph in  $(CCC_n^{ab})'$ . Then we replace the edge  $((n-3, g(\mathbf{x})), (0, g(\mathbf{x})))$  with the path  $\langle (n-3, g(\mathbf{x})), (n-2, g(\mathbf{x})), (n-1, g(\mathbf{x})), (0, g(\mathbf{x})) \rangle$  if the edge  $((n-3, g(\mathbf{x})), (0, g(\mathbf{x})))$  is in  $F$  to obtain a subgraph of  $CCC_n^{ab}$ . The resultant graph is called  $G^{ab}$ .

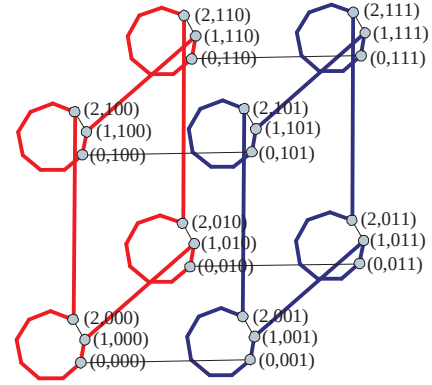
## 3 Two basic subgraphs of $CCC_n$

Now, we recursively construct two subgraphs of  $CCC_n$ ,  $X_n$  and  $Y_n$  for odd integer  $n$  with  $n \geq 3$  as follows:

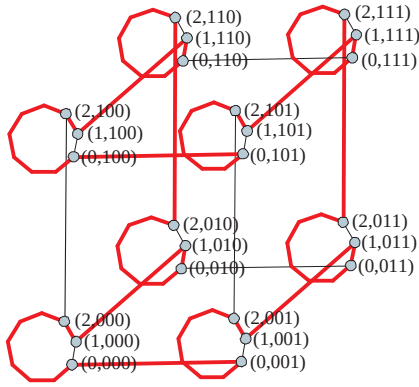
Let  $X_3$  be the spanning subgraph of  $CCC_3$  with edge set  $\{((0, 000), (1, 000)), ((1, 000), (1, 010)), ((1, 010), (0, 010)), ((0, 010), (2, 010)), ((2, 010), (2, 110)), ((2, 110), (0, 110)), ((0, 110), (1, 110)), ((1, 110), (1, 100)), ((1, 100), (0, 100)), ((0, 100), (2, 100)), ((2, 100), (2, 000)), ((2, 000), (0, 000)), ((0, 001), (1, 001)), ((1, 001), (1, 011)), ((1, 011), (0, 011)), ((0, 011), (2, 011)), ((2, 011), (2, 111)), ((2, 111), (0, 111)), ((0, 111), (1, 111)), ((1, 111),$

$(1, 101)), ((1, 101), (0, 101)), ((0, 101), (2, 101)), ((2, 101), (2, 001)), ((2, 001), (0, 001))\}$ . See Figure 1 for illustration. It is easy to observe that  $X_3$  forms two disjoint cycles such that one cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is a 3-bit binary string such that } (\mathbf{x})_0 = 0\}$  and the other cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is a 3-bit binary string such that } (\mathbf{x})_0 = 1\}$ . Moreover,  $((2, \mathbf{x}), (0, \mathbf{x})) \in E(X_3)$  for every 3-bit binary string  $\mathbf{x}$ . Assume that  $X_{n-2}$  be the spanning subgraph of  $CCC_{n-2}$ .  $X_{n-2}$  forms two disjoint cycles such that one cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is an } (n-2)\text{-bit binary string such that } (\mathbf{x})_0 = 0\}$ , the other cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is an } (n-2)\text{-bit binary string such that } (\mathbf{x})_0 = 1\}$ . Moreover,  $((n-3, \mathbf{x}), (0, \mathbf{x})) \in E(X_{n-2})$  for every  $(n-2)$ -binary string  $\mathbf{x}$ . Now, we construct a spanning subgraph  $X_n$  of  $CCC_n$  for  $n \geq 5$  with edge set  $E(X_{n-2}^{00}) \cup E(X_{n-2}^{01}) \cup E(X_{n-2}^{11}) \cup E(X_{n-2}^{10})$ . It is easy to observe that  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(X_{n-2}^{ab})$  for every  $n$ -binary string  $\mathbf{x}$  and  $ab \in \{00, 01, 10, 11\}$ . Then, we delete the edge set  $\{((n-1, ab0^{n-3}0), (n-2, ab0^{n-3}0)), ((n-1, ab0^{n-3}1), (n-2, ab0^{n-3}1)) \mid ab \in \{00, 01, 10, 11\}\}$  and add the edge set  $\{((n-2, 000^{n-3}0), (n-2, 010^{n-3}0)), ((n-2, 000^{n-3}1), (n-2, 010^{n-3}1)), ((n-2, 100^{n-3}0), (n-2, 110^{n-3}0)), ((n-2, 100^{n-3}1), (n-2, 110^{n-3}1)), ((n-1, 000^{n-3}0), (n-1, 100^{n-3}0)), ((n-1, 000^{n-3}1), (n-1, 100^{n-3}1)), ((n-1, 010^{n-3}0), (n-1, 110^{n-3}0)), ((n-1, 010^{n-3}1), (n-1, 110^{n-3}1))\}$ . The resultant graph is  $X_n$ . Obviously, that  $X_n$  forms two disjoint cycles such that one cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is an } n\text{-bit binary string such that } (\mathbf{x})_0 = 0\}$  and the other cycle spans the set  $\{(l, \mathbf{x}) \mid \mathbf{x} \text{ is an } n\text{-bit binary string such that } (\mathbf{x})_0 = 1\}$ . Moreover,  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(X_n)$  for every  $n$ -binary string  $\mathbf{x}$ .

Let  $Y_3$  be the spanning subgraph of  $CCC_3$  with edge set  $\{((1, 000), (1, 010)), ((1, 010), (0, 010)), ((0, 010), (2, 010)), ((2, 010), (2, 110)), ((2, 110), (0, 110)), ((0, 110), (1, 110)), ((1, 110), (1, 100)), ((1, 100), (2, 100)), ((2, 100), (0, 100)), ((0, 100), (0, 010)), ((0, 010), (2, 010)), ((2, 010), (1, 101)), ((1, 101), (1, 111)), ((1, 111), (0, 111)), ((0, 111), (2, 111)), ((2, 111), (2, 011)), ((2, 011), (0, 011)),$

Figure 1: The graph  $X_3$ 

$((0, 011), (1, 011)), ((1, 011), (1, 001)), ((1, 001), (2, 001)), ((2, 001), (0, 001)), ((0, 001), (0, 000)), ((0, 000), (2, 000)), ((2, 000), (1, 000))\}$ . See Figure 2 for illustration. It is easy to observe that  $Y_3$  forms a hamiltonian cycle of  $CCC_3$ . Moreover,  $((2, \mathbf{x}), (0, \mathbf{x})) \in E(Y_3)$  for every 3-bit binary string  $\mathbf{x}$ . Assume that  $Y_{n-2}$  be the spanning subgraph of  $CCC_{n-2}$ .  $Y_{n-2}$  forms a hamiltonian cycle of  $CCC_{n-2}$ . Moreover,  $((n-3, \mathbf{x}), (0, \mathbf{x})) \in E(Y_{n-2})$  for every  $(n-2)$ -bit binary string  $\mathbf{x}$ . Now, we construct a spanning subgraph of  $CCC_n$  for  $n \geq 5$  with edge set  $E(Y_{n-2}^{00}) \cup E(X_{n-2}^{01}) \cup E(X_{n-2}^{10}) \cup E(X_{n-2}^{11})$ . It is easy to observe that  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(Y_{n-2}^{00}, E(X_{n-2}^{01}), E(X_{n-2}^{10}), E(X_{n-2}^{11})$  for every  $n$ -bit binary string  $\mathbf{x}$ . Then, we delete the edge set  $\{((n-1, ab0^{n-3}0), (n-2, ab0^{n-3}0)), ((n-1, ab0^{n-3}1), (n-2, ab0^{n-3}1)) \mid ab \in \{00, 01, 10, 11\}\}$  and add the edge set  $\{((n-2, 000^{n-3}0), (n-2, 010^{n-3}0)), ((n-2, 000^{n-3}1), (n-2, 010^{n-3}1)), ((n-2, 100^{n-3}0), (n-2, 110^{n-3}0)), ((n-2, 100^{n-3}1), (n-2, 110^{n-3}1)), ((n-1, 000^{n-3}0), (n-1, 100^{n-3}0)), ((n-1, 000^{n-3}1), (n-1, 100^{n-3}1)), ((n-1, 010^{n-3}0), (n-1, 110^{n-3}0)), ((n-1, 010^{n-3}1), (n-1, 110^{n-3}1))\}$ . The resultant graph is  $Y_n$ . Obviously, that  $Y_n$  forms a hamiltonian cycle of  $CCC_n$ . Moreover,  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(Y_n)$  for every  $n$ -bit binary string  $\mathbf{x}$ .

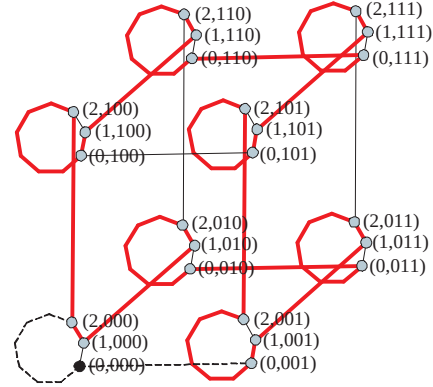
Figure 2: The graph  $Y_3$ 

## 4 Main Result

**Lemma 4.1** *Let  $n$  be an odd integer and  $n \geq 3$ . There exists a hamiltonian cycle  $H_n$  in  $CCC_n - \{(0, \mathbf{0}^n)\}$  and edge  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in H_n$  for every nonzero  $n$ -bit binary string  $\mathbf{x}$ .*

**Proof.** Let  $H_3$  be the spanning subgraph of  $CCC_3 - \{(0, 000)\}$  with edge set  $\{((1, 000), (1, 010)), ((1, 010), (2, 010)), ((2, 010), (0, 010)), ((0, 010), (0, 011)), ((0, 011), (2, 011)), ((2, 011), (1, 011)), ((1, 011), (1, 001)), ((1, 001), (0, 001)), ((0, 001), (2, 001)), ((2, 001), (2, 101)), ((2, 101), (0, 101)), ((0, 101), (1, 101)), ((1, 101), (1, 111)), ((1, 111), (2, 111)), ((2, 111), (0, 111)), ((0, 111), (0, 110)), ((0, 110), (2, 110)), ((2, 110), (1, 110)), ((1, 110), (1, 100)), ((1, 100), (0, 100)), ((0, 100), (2, 100)), ((2, 100), (2, 000)), ((2, 000), (1, 000))\}$ . See Figure 3 for illustration. It is easy to observe that  $H_3$  forms a hamiltonian cycle of  $CCC_3 - (0, \mathbf{0}^3)$ . Moreover,  $((2, \mathbf{x}), (0, \mathbf{x})) \in E(H_3)$  for every nonzero 3-bit binary string  $\mathbf{x}$ . Assume that  $H_{n-2}$  be the hamiltonian cycle of  $CCC_{n-2} - \{(0, \mathbf{0}^{n-2})\}$ . Moreover,  $((n-3, \mathbf{x}), (0, \mathbf{x})) \in E(H_{n-2})$  for every nonzero  $(n-2)$ -bit binary string  $\mathbf{x}$ . Now, we construct a hamiltonian cycle of  $CCC_n$  for  $n \geq 5$  with edge set  $E(H_{n-2}^{00}) \cup E(X_{n-2}^{01}) \cup E(X_{n-2}^{11}) \cup E(Y_{n-2}^{10})$ . It is easy to observe that  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(H_{n-2}^{00}, E(X_{n-2}^{01}), E(Y_{n-2}^{10}), E(X_{n-2}^{11}))$  for every nonzero  $n$ -bit binary string  $\mathbf{x}$ . Then, we delete the edge set  $\{((n-1, 000^{n-3}1), (n-2, 000^{n-3}1))\} \cup \{((n-1, ab0^{n-3}0), (n-2, ab0^{n-3}0)), ((n-$

$1, ab0^{n-3}1), (n-2, ab0^{n-3}1)) \mid ab \in \{01, 10, 11\}\}$  and add the edge set  $\{((n-2, 000^{n-3}0), (n-2, 010^{n-3}0)), ((n-2, 000^{n-3}1), (n-2, 010^{n-3}1)), ((n-2, 100^{n-3}0), (n-2, 110^{n-3}0)), ((n-2, 100^{n-3}1), (n-2, 110^{n-3}1)), ((n-1, 000^{n-3}0), (n-1, 100^{n-3}0)), ((n-1, 000^{n-3}1), (n-1, 100^{n-3}1)), ((n-1, 010^{n-3}0), (n-1, 110^{n-3}0)), ((n-1, 010^{n-3}1), (n-1, 110^{n-3}1))\}$ . The resultant graph is  $H_n$ . Obviously, that  $H_n$  forms a hamiltonian cycle of  $CCC_n - \{(0, \mathbf{0}^n)\}$ . Moreover,  $((n-1, \mathbf{x}), (0, \mathbf{x})) \in E(H_n)$  for every nonzero  $n$ -bit binary string  $\mathbf{x}$ .

Figure 3: The graph  $H_3$ 

We know that cube-connected cycles graph  $CCC_n$  is vertex symmetric. By Lemma 4.1, we have the following main theorem.

**Theorem 4.1** *The cube-connected cycles graph  $CCC_n$  is 1-vertex-fault-tolerant hamiltonian when  $n$  is odd and  $n \geq 3$ .*

## References

- [1] F. Annexstein, M. Baumslag, A.L. Rosenberg, Group action graphs and parallel architectures, SIAM Journal of Computers 19 (1990) 544–569.
- [2] J. Bruck, R. Cypher, and C.T. Ho, On the construction of fault-tolerant cube-connected cycles networks, Journal of Parallel and Distributed Computing, 25 (1995) 98–106.

- [3] T. Dvořák and P. Gregor, Hamiltonian fault-tolerance of hypercubes, *Electronic Notes in Discrete Mathematics*, 29 (2007) 471–477.
- [4] I. Friš, I. Havel, and P. Liebl, The diameter of the cube-connected cycles, *Information Processing Letters*, 61 (1997) 157–160.
- [5] A. Germa, M.C. Heydemann, D. Sotteau, Cycles in the cube-connected cycles graph, *Discrete Applied Mathematics* 83 (1998) 135–155.
- [6] S.Y. Hsieh, C.N. Kuo, and H.L. Huang, 1-vertex-fault-tolerant cycles embedding on folded hypercubes, *Discrete Applied Mathematics*, 157 (2009) 3094–3098.
- [7] L.H. Hsu, C.K. Lin, *Graph Theory and Interconnection Networks*, CRC Press, 2008.
- [8] F.P. Preparata, V. Vuillemin, The cube-connected cycles: a versatile network for parallel computation, *commun. of the ACM* 24 (1981) 300–309.
- [9] A.L. Rosenberg, *Cycles in networks*, Tech. Report UM-CS-1991-20, Univ. of Massachusetts, 1991.
- [10] R. Stong, On hamiltonian cycles in Cayley graphs of wreath products, *Discrete Mathematics* 85 (1987) 75–80.